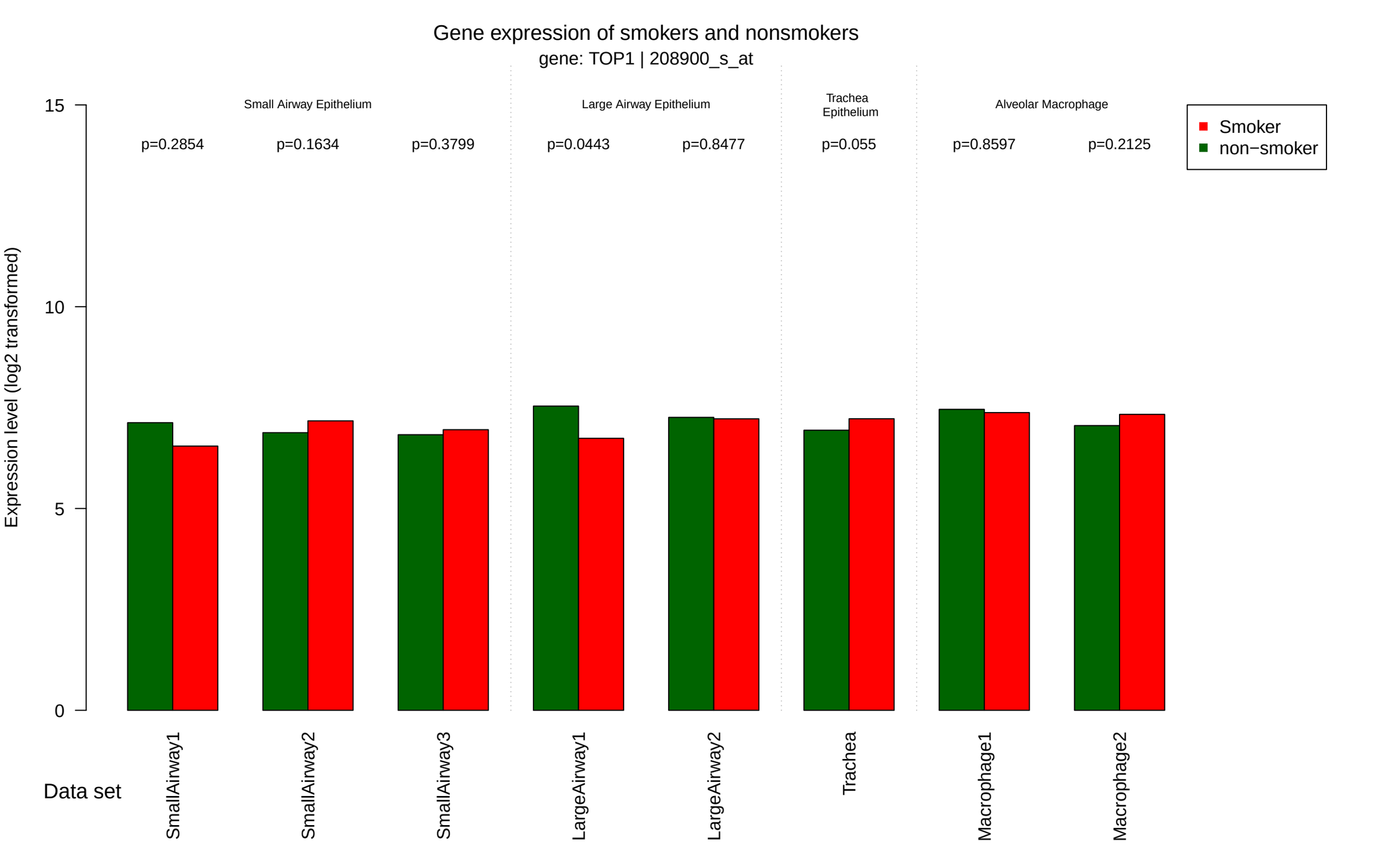
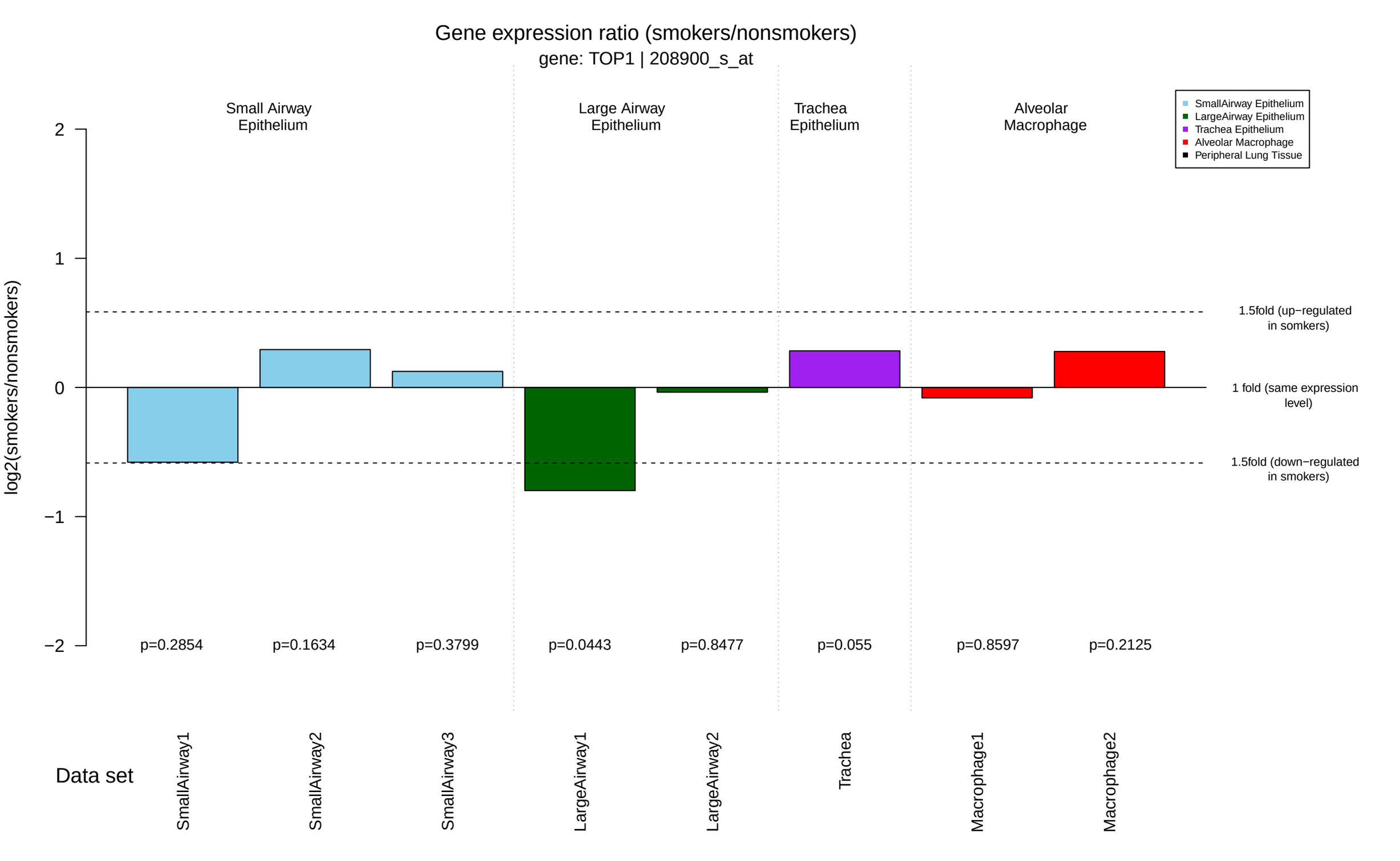
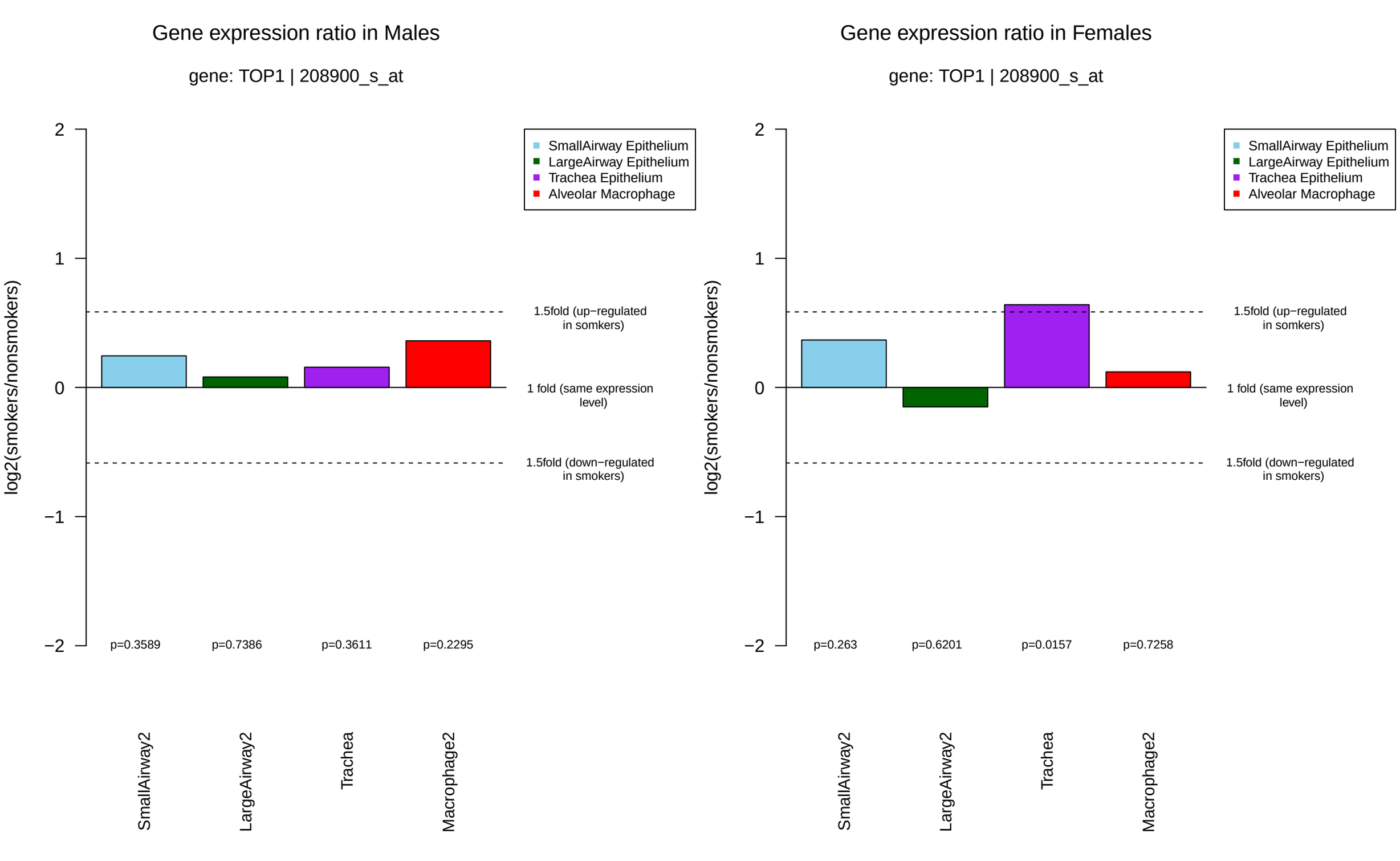
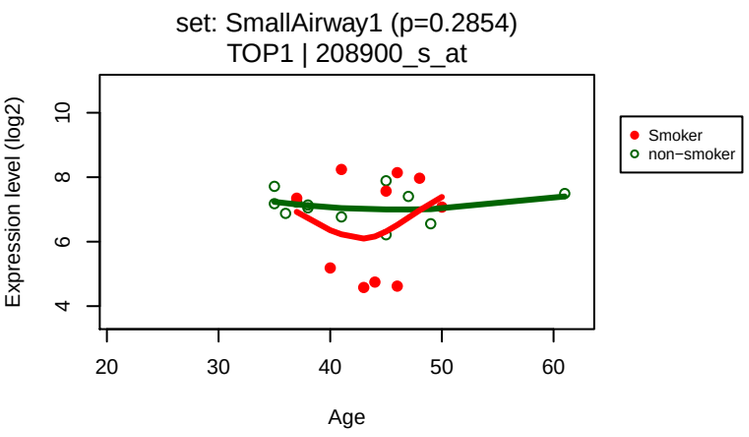
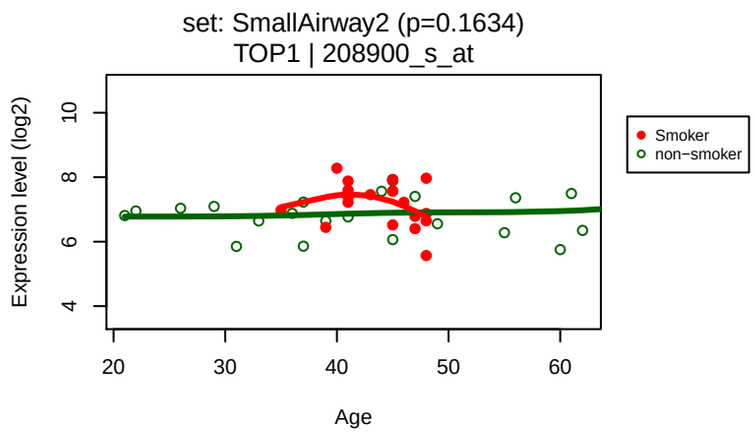
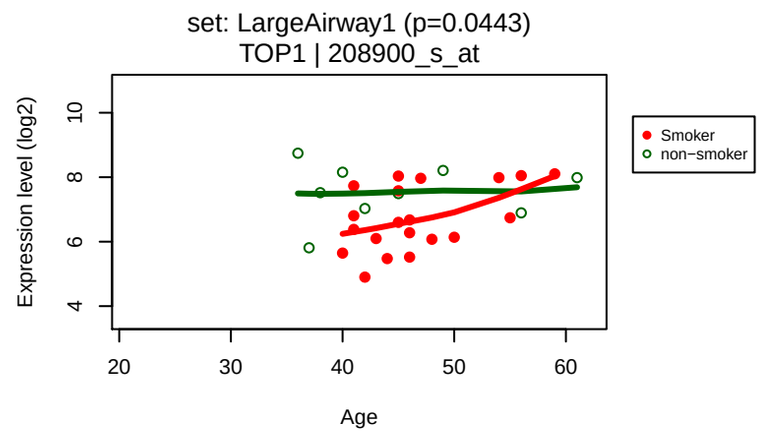

Gene expression of smokers and nonsmokers
gene: TOP1 | 208900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2854
p=0.1634
p=0.3799
p=0.0443
p=0.8477
p=0.055
p=0.8597
p=0.2125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

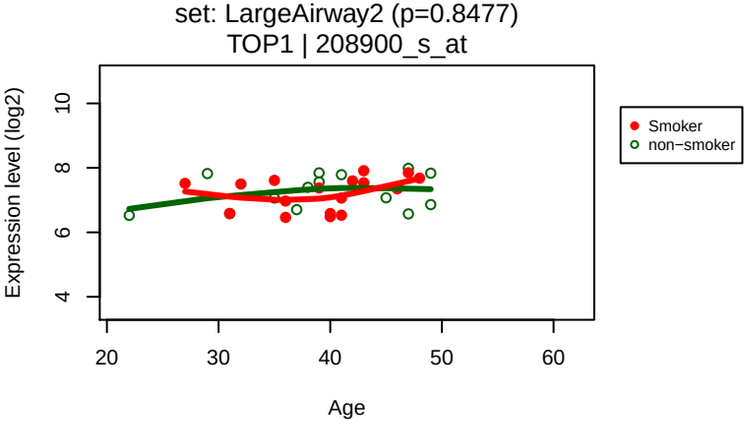

Gene expression ratio (smokers/nonsmokers)
gene: TOP1 | 208900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2854
p=0.1634
p=0.3799
p=0.0443
p=0.8477
p=0.055
p=0.8597
p=0.2125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

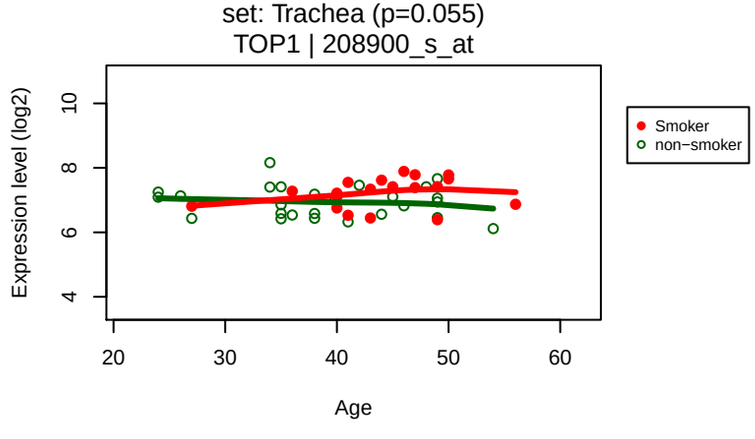

Gene expression ratio in Males
gene: TOP1 | 208900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3589
p=0.7386
p=0.3611
p=0.2295
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TOP1 | 208900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.263
p=0.6201
p=0.0157
p=0.7258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

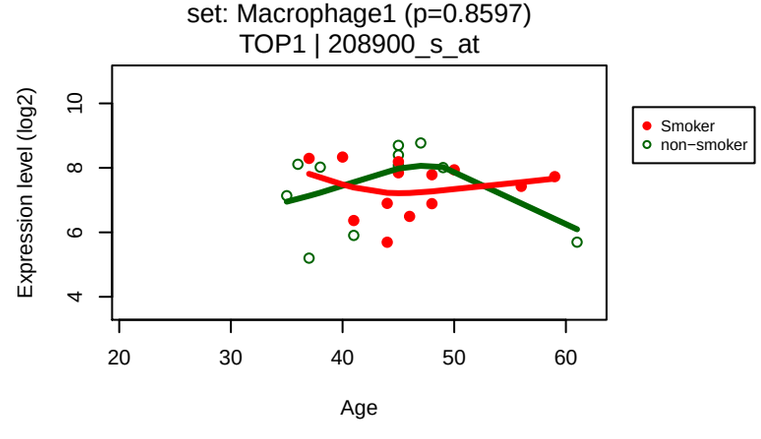

set: SmallAirway1 (p=0.2854)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker

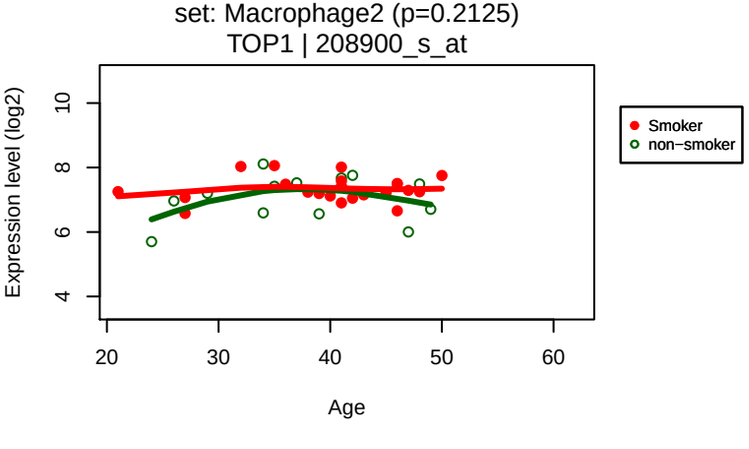

set: SmallAirway2 (p=0.1634)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker

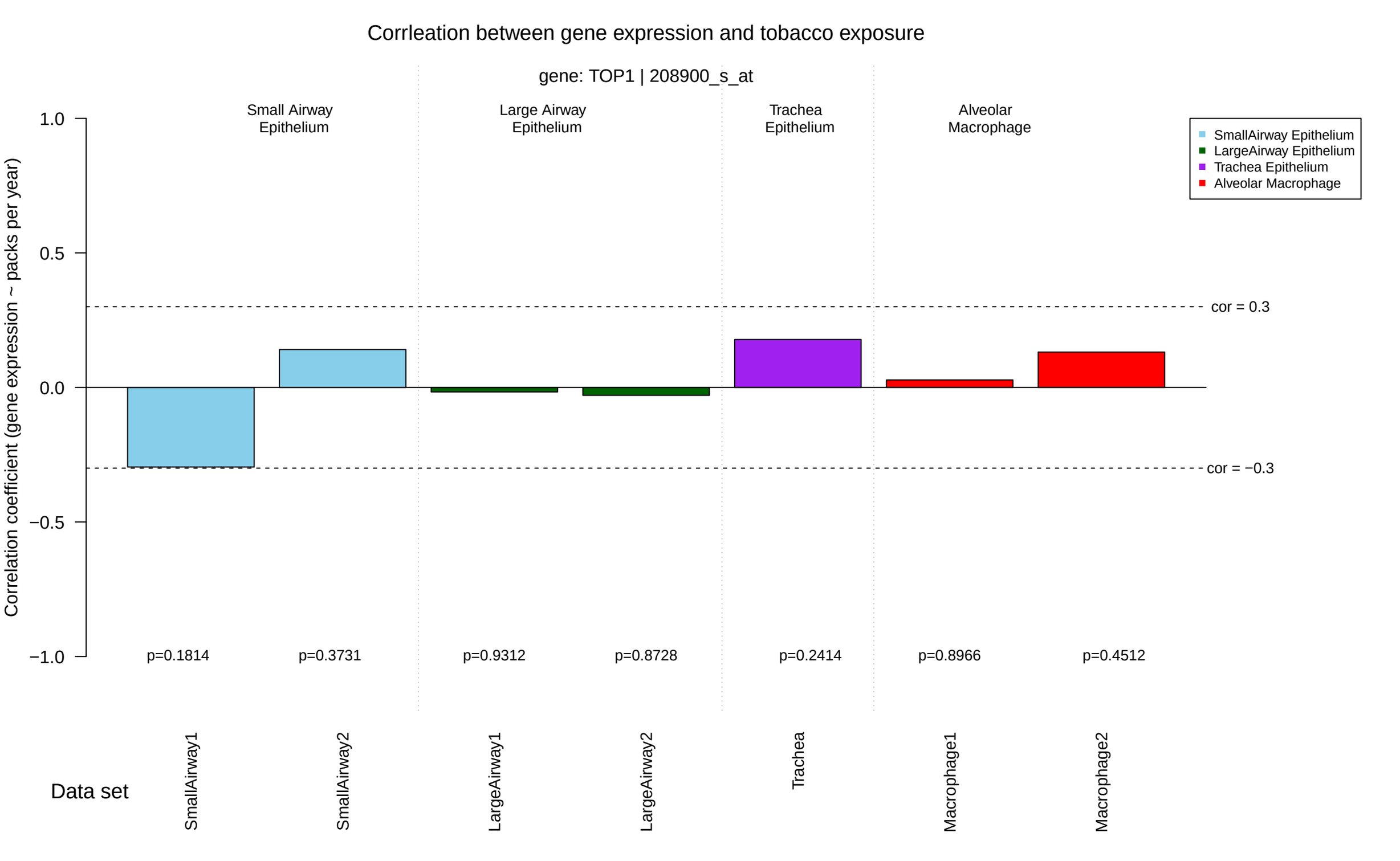

set: LargeAirway1 (p=0.0443)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8477)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.055)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8597)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2125)
TOP1 | 208900_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TOP1 | 208900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1814
p=0.3731
p=0.9312
p=0.8728
p=0.2414
p=0.8966
p=0.4512
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2